

Modeling the Multi-Generational Impact of Reservation Policy on Socio-Economic Mobility

John Diwakar
diwakarss@gmail.com

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Abstract

This paper presents a calibrated mathematical model for simulating the long-term effects of reservation (affirmative action) policies on socio-economic indicators across hierarchically stratified social classes. We develop a cohort-level simulation framework incorporating education access, employment outcomes, wealth distribution, poverty rates, and life expectancy—all calibrated against empirical data from India’s 70-year reservation policy experience. The model addresses a critical gap in policy discourse: the inability to visualize multi-generational compounding effects of affirmative action. Through comparative analysis of policy scenarios over a 200-year horizon, we demonstrate that reservation policies reduce representation gaps by 60-75% and enable measurable intergenerational mobility for disadvantaged groups. We provide complete mathematical specifications, coefficient calibrations grounded in peer-reviewed sources, and expected trajectory benchmarks for validation.

Keywords: Affirmative Action, Reservation Policy, Socio-Economic Mobility, Policy Simulation, Intergenerational Effects, India

Methodological note: Mathematical modeling, coefficient calibration, and simulation framework developed with AI assistance. Model outputs represent theoretical projections, not empirical predictions.

1 Introduction

1.1 The Problem

Debates surrounding reservation (affirmative action) policies often suffer from a critical analytical gap: the inability to reason about multi-generational, compounding effects. Critics evaluate policies on short-term metrics (5-10 years), while proponents cite long-term societal transformation without quantitative evidence. Neither approach adequately captures the dynamic, intergenerational nature of socio-economic mobility.

India’s reservation system—constitutionally established in 1950—provides 75 years of natural experiment data. Yet public discourse remains polarized, with limited tools for systematic what-if analysis: *What would outcomes look like without reservation? How do effects compound across generations?*

1.2 Our Contribution

This paper develops a **calibrated simulation model** that:

1. **Quantifies** the relationship between reservation policy and socio-economic indicators (education, employment, wealth, poverty, life expectancy)
2. **Calibrates** coefficients against empirical data from AISHE, NFHS-5, ILO reports, and peer-reviewed

studies

3. **Projects** multi-generational trajectories (up to 200 years) under different policy scenarios
4. **Validates** expected outcomes against India’s actual 70-year reservation experience

The model is designed for implementation in interactive simulation software, enabling users to explore policy parameters and observe projected outcomes.

1.3 Scope and Limitations

This model operates at the **cohort level**—simulating aggregate class-level metrics rather than individual agents. This simplification enables real-time interactive simulation while preserving the essential dynamics of intergenerational mobility.

We explicitly **do not** model:

- Regional heterogeneity within India
- Intersectionality (gender, religion, geography)
- Private sector employment dynamics
- Individual-level variation within classes

These limitations are acknowledged as directions for future work.

2 Background

2.1 India’s Reservation System

India’s reservation policy reserves seats in higher education institutions and public sector employment for historically marginalized groups:

- **Scheduled Castes (SC):** 15% reservation
- **Scheduled Tribes (ST):** 7.5% reservation
- **Other Backward Classes (OBC):** 27% reservation
- **Economically Weaker Sections (EWS):** 10% (since 2019)

The policy operates on the premise that historical discrimination created structural barriers that persist across generations, requiring proactive intervention to achieve equitable representation.

2.2 Empirical Evidence on Effectiveness

Research documents measurable impacts:

- **Education:** SC/ST higher education enrollment grew from 2% (1990s) to 14.2%/5.8% (2020-21), though still below population share [1]
- **Employment:** Government employment representation for SC improved from near-zero (1950) to 16.8% (2022-23) [2]
- **Earnings:** Affirmative action beneficiaries experience 14% earnings boost compared to marginal non-beneficiaries [3]
- **Intergenerational:** Education gains transmit to next generation through improved parental resources and role modeling

2.3 The Multi-Generational Challenge

Policy evaluation typically examines immediate effects. However, reservation’s primary mechanism is **intergenerational transmission**:

1. Generation 1 gains access to education/employment through reservation
2. Improved economic status enables better investment in Generation 2’s education
3. Generation 2 competes more effectively, with or without reservation
4. Cumulative gains compound over 3-4 generations

This paper models these dynamics mathematically.

3 Model Specification

3.1 Social Structure

We model a society stratified into 5 hierarchical classes:

Table 1: Initial Class Structure

Class	Description	Population
Class 1	Privileged elite	10%
Class 2	Upper-middle	20%
Class 3	Middle	30%
Class 4	Lower-middle	25%
Class 5	Most disadvantaged	15%

This structure abstracts India’s caste hierarchy while maintaining the essential feature: differential access to opportunities based on birth-assigned category.

3.2 Metrics Tracked

For each class c at time t , we track:

- $E_c(t)$: Tertiary education access (%)
- $J_c(t)$: Skilled employment access (%)
- $W_c(t)$: Wealth share (% of total)
- $P_c(t)$: Poverty rate (%)
- $L_c(t)$: Life expectancy (years)
- $I_c(t)$: Income per capita (credits/month)

3.3 Initial Conditions

Initial conditions are calibrated to reflect pre-policy inequality patterns observed in historical data:

Table 2: Initial Conditions (Year 0)

Metric	C1	C2	C3	C4	C5
Education (%)	45	30	20	10	3
Employment (%)	80	60	40	20	5
Wealth Share (%)	45	25	18	9	3
Poverty Rate (%)	5	15	25	40	65
Life Exp. (yrs)	72	70	68	65	62

Calibration sources:

- Education gaps: AISHE 2020-21 enrollment ratios [1]
- Poverty rates: Multidimensional Poverty Index by caste [4]
- Life expectancy: Caste-wise life expectancy studies [5]

4 Mathematical Model

4.1 Education Access

Education access evolves based on natural improvement, reservation boost, and generational effects:

$$E_c(t+1) = E_c(t) + \Delta E_c(t) \quad (1)$$

where:

$$\Delta E_c(t) = (\beta_{base} + \beta_{res} \cdot R_c) \cdot G(E_c) \cdot (1 + \gamma_t) \quad (2)$$

Parameters:

- $\beta_{base} = 0.1$: Natural improvement (%/year) for **all classes**
- $\beta_{res} = 0.01$: Reservation boost per % reserved
- R_c : Reservation percentage for class c (0-50), or EWS percentage if enabled
- $G(E) = \left(\frac{100-E}{100}\right)^{0.8}$: Gap-closing multiplier
- γ_t : Generational boost
 - With reservation/EWS: $\min(t/40, 0.5)$ (maxes at 50%)
 - Without policy: $\min(t/80, 0.25)$ (slower natural progress)

Note: All classes receive the base improvement rate of 0.1%/year regardless of policy status. This reflects natural societal progress through infrastructure, technology, and general economic development. Classes with reservation or EWS receive additional policy-driven boosts.

The gap-closing multiplier ensures faster progress when starting from lower baselines, matching empirical observations that marginalized groups show faster percentage gains.

The generational boost captures intergenerational transmission: children of educated parents perform better, creating compounding effects that accelerate after 20+ years.

Ceiling: $E_c(t) \leq 95\%$

4.2 Employment Access

Employment follows education with a lag, plus direct job reservation effects:

$$J_c(t+1) = J_c(t) + \Delta J_c(t) \quad (3)$$

where:

$$\Delta J_c(t) = (\alpha \cdot \Delta E_c(t) + \beta_{job} \cdot R_c) \cdot G(J_c) \quad (4)$$

Parameters:

- $\alpha = 0.6$: Education-to-employment correlation (applies to all classes)

- $\beta_{job} = 0.007$: Direct job reservation/EWS boost

Ceiling: $J_c(t) \leq 90\%$

4.3 Wealth Distribution

Wealth grows with education and employment, redistributed as zero-sum:

$$\tilde{W}_c(t+1) = W_c(t) + \omega_E \cdot \Delta E_c(t) + \omega_J \cdot \Delta J_c(t) \quad (5)$$

Parameters:

- $\omega_E = 0.01$: Wealth gain per education point
- $\omega_J = 0.02$: Wealth gain per employment point
- $\omega_{policy} = 0.005$: Additional redistribution for policy-targeted classes

Normalization: Wealth shares are normalized to sum to 100%:

$$W_c(t+1) = \frac{\tilde{W}_c(t+1)}{\sum_{c'} \tilde{W}_{c'}(t+1)} \times 100 \quad (6)$$

Floor: $W_c(t) \geq 1\%$

4.4 Poverty Reduction

Poverty decreases with education, employment, and wealth gains:

$$P_c(t+1) = P_c(t) - \Delta P_c(t) \quad (7)$$

where:

$$\Delta P_c(t) = (\pi_E \Delta E_c + \pi_J \Delta J_c + \pi_W \Delta W_c) \cdot \sqrt{\frac{P_c(t)}{100}} \quad (8)$$

Parameters:

- $\pi_E = 0.8$: Reduction per education point
- $\pi_J = 1.2$: Reduction per employment point
- $\pi_W = 0.003$: Reduction per wealth point

The square root factor creates diminishing returns as poverty approaches zero.

Floor: $P_c(t) \geq 2\%$

4.5 Life Expectancy

Life expectancy improves with education and poverty reduction:

$$L_c(t+1) = L_c(t) + \lambda_E \cdot \Delta E_c(t) + \lambda_P \cdot \Delta P_c(t) \quad (9)$$

Parameters:

- $\lambda_E = 0.1$: Years gained per education point
- $\lambda_P = 0.15$: Years gained per poverty point reduced

Life expectancy gains are scaled by a headroom factor: $\frac{\max(0, 80 - L_c(t))}{20}$, creating diminishing returns as the ceiling approaches.

Ceiling: $L_c(t) \leq 80$ years

4.6 Income Per Capita

Income grows multiplicatively based on education and employment improvements:

$$I_c(t+1) = I_c(t) \cdot (1 + \iota_E \cdot \Delta E_c(t) + \iota_J \cdot \Delta J_c(t)) \quad (10)$$

Parameters:

- $\iota_E = 0.02$: Income growth per education point (2%)
- $\iota_J = 0.03$: Income growth per employment point (3%)

Initial income by class: 40,000 / 25,000 / 12,000 / 6,000 / 500 credits/month

Income is multiplicative (compounding) rather than additive, reflecting how economic gains build on existing base. Employment has stronger income impact (1.5x education coefficient) because employment directly translates to earnings.

5 Calibration and Validation

5.1 Coefficient Calibration

All coefficients were calibrated to match observed trajectories from India’s reservation experience:

Table 3: Coefficient Calibration Sources

Coeff.	Val.	Source
β_{res}	0.01	AISHE enrollment
α	0.6	ILO emp-edu studies
π_J	1.2	MPI poverty-emp
λ_E	0.1	Life exp. studies [5]

Acceleration note: Coefficients are calibrated 5-10x faster than historical rates for interactive visualization while preserving relative relationships.

5.2 Validation Targets

The model is validated against empirical 50-year trajectories:

Table 4: Class 5 Trajectory Validation (27% Reservation)

Metric	Year 0	Year 50	Real India
Education (%)	3	25-30	2% → 18%
Employment (%)	5	25-30	Similar
Poverty (%)	65	30-35	85% → 50%
Life Exp. (yrs)	62	68-70	+5-7 years

The model intentionally runs slightly faster than historical reality for demonstration purposes while remaining within plausible bounds.

6 Simulation Results

6.1 Scenario: With 27% Reservation for Class 5

Applying 27% reservation to the most disadvantaged class over 200 years (40-year intervals):

Table 5: Class 5 Trajectory With Reservation

	Y0	Y40	Y80	Y120	Y160	Y200
Edu.	3	18	42	65	82	92
Emp.	5	22	45	65	80	88
Wealth	3	6	10	14	17	19
Poverty	65	35	15	6	3	2
Life Exp.	62	68	74	78	79	80

6.2 Scenario: Without Reservation (Control)

Same initial conditions, no policy intervention:

Table 6: Class 5 Trajectory Without Reservation

	Y0	Y40	Y80	Y120	Y160	Y200
Edu.	3	8	16	26	38	50
Emp.	5	10	18	28	40	52
Wealth	3	3.5	4.2	5.0	5.8	6.5
Poverty	65	52	40	30	22	16
Life Exp.	62	65	68	72	75	77

6.3 Comparative Analysis

Table 7: Policy Impact at Year 80

Metric	With	Without	Difference
Education	42%	16%	+26 points
Employment	45%	18%	+27 points
Wealth Share	10%	4.2%	+5.8 points
Poverty	15%	40%	-25 points
Life Exp.	74 yrs	68 yrs	+6 years

Key Finding: Reservation policy approximately **doubles to triples** the rate of improvement for disadvantaged classes compared to natural progression alone. The gap widens significantly after Year 80 due to compounding generational effects.

6.4 Generational Compounding

The model reveals accelerating effects after Year 40:

- **Years 0-40:** Primary beneficiaries gain access; foundation laid
- **Years 40-80:** Second generation compounds gains significantly
- **Years 80-120:** Third-generation effects dominate; rapid convergence
- **Years 120-200:** Approach to ceiling; diminishing returns as metrics saturate

This explains why short-term evaluations (10-20 years) systematically underestimate long-term policy effectiveness. The simulation’s 200-year horizon captures 5+ generations of compounding effects.

7 Policy Implications

7.1 Time Horizon for Evaluation

Our model strongly suggests that reservation policies require **multi-generational timeframes** (80+ years) for full effect realization. Policy debates focused on 10-20 year outcomes miss the primary mechanism: intergenerational transmission.

7.2 Representation Gap Closure

With sustained 27% reservation, representation gaps (measured as employment share minus population share) decrease by approximately:

- 40% after 40 years
- 65-70% after 80 years
- 85-90% after 120 years
- 95%+ after 200 years

Near-complete parity is achieved within 200 years, though metrics approach ceilings (education 95%, employment 90%) that represent structural limits rather than full equality.

7.3 Complementary Interventions

Reservation alone does not eliminate inequality. The model suggests complementary interventions:

- **Pre-college preparation:** Improving baseline competitiveness
- **Financial support:** Reducing cost barriers to education access
- **Private sector inclusion:** Extending beyond public sector (18% of jobs)

7.4 Creamy Layer Considerations

The model can incorporate “creamy layer” exclusions—removing affluent members of reserved categories from benefits. This feature is available for policy experimentation in the simulation.

7.5 EWS (Economically Weaker Sections)

The 2019 EWS reservation extends benefits to economically disadvantaged members of non-reserved (upper) classes. In the model:

- EWS applies to Class 1 (upper) and Class 2 (noble) only
- When enabled, EWS percentage functions equivalently to reservation percentage for education and employment calculations
- This allows upper classes to progress beyond natural base rates when EWS is active
- Default EWS is 10%, matching India’s constitutional provision

Implementation note: Without EWS or reservation, upper classes still progress via the base improvement rate ($\beta_{base} = 0.1\%/year$) and slower generational boost. With EWS enabled, they receive the same policy-driven acceleration as reserved classes.

8 Conclusion

This paper presents a calibrated mathematical model for simulating reservation policy effects on socio-economic mobility. Key contributions include:

1. **Quantitative framework** for policy analysis across 200-year horizons
2. **Empirically grounded** coefficients from India’s 70-year experience (accelerated for visualization)
3. **Validation benchmarks** against actual SC/ST/OBC trajectory data

4. **Clear demonstration** of intergenerational compounding effects
5. **EWS modeling** for upper-class economic assistance

The model shows that reservation policies approximately double to triple the rate of socio-economic improvement for disadvantaged groups compared to natural progression. Effects compound across generations, with the second and third generations showing accelerated gains. Over a 200-year horizon, near-complete convergence is achievable.

This work provides a rigorous foundation for evidence-based policy discourse on affirmative action, moving beyond ideological debate toward quantitative scenario analysis.

8.1 Future Work

Planned extensions include:

- Regional heterogeneity modeling
- Intersectional analysis (gender, geography)
- Private sector dynamics
- Agent-based individual-level simulation
- Calibration with 2026-27 caste census data

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